

```

        return -EIO;
    }
}

phys = fe.blocks[iblock];
map_bh(bh_result, sb, phys);

return 0;
}

```

- 4) The *open*, *release*, *read*, *write*, *aio_read*, *aio_write* and *fsync* (under a regular file's struct *file_operations*) should basically go through the buffer cache, i.e., they should be implemented using the address-space operations. This being a common requirement, the kernel provides a generic set of kernel APIs, namely *generic_file_open()*, *do_sync_read()*, *do_sync_write()*, *generic_file_aio_read()*, *generic_file_aio_write()*, and *simple_sync_file()/noop_fsync()*. Note that there is no API for *release*, as it is a 'return 0' API. Check out the *real_sfs.c* file for the exact definition of struct *file_operations* *sfs_fops*.
- 5) As for *readdir* (under a directory's struct *file_operations*): *sfs_readdir()* primarily reads the directory entries of an underlying directory (denoted by *file*), and fills it up into the VFS directory entry cache (pointed to by *dentry*) using the parameter function *filldir*. As real SFS has only one directory (the top one), this function basically fills up the directory entry cache with directory entries for all the

files in the underlying file system, using the transformed *sfs_list()* (defined in *real_sfs_ops.c*), adapted from the *browse_real_sfs* application. Check out the *real_sfs.c* file for the complete *sfs_readdir()* implementation, which starts with filling directory entries for '.' (current directory) and '..' (parent directory), and then for all the files of the real SFS, using *sfs_list()*.

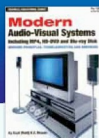
With all these in place, one can see the amazing demo by Pugs, as shown in Figure 1. And don't forget to watch the live log in */var/log/messages* using a *'tail -f /var/log/messages'*, and to match it with every command you issue on the mounted real SFS file system. This would give you the best insight into when each system call gets invoked. Or, in other words, which application invokes which system call from the file system front. To trace all the system calls invoked by an application/command, use *strace* with the command; e.g., type *'strace ls'* instead of just *'ls'*. **END** 

By: Anil Kumar Pugalia

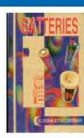
The author is a freelance trainer in Linux internals, Linux device drivers, embedded Linux and related topics. Prior to this, he was at Intel and Nvidia. He has been working with Linux since 1994. A gold medalist from the Indian Institute of Science, Linux and knowledge sharing are two of his many passions. Creating and playing around with open source hardware is one of his hobbies. Learn more about him at <http://profession.sanika-pugs.com>, including information about his open source hardware freak-outs and his various Linux related training sessions and workshops. He can be reached at email@sanika-pugs.com.

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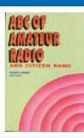
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